Submission 2: Plotting Function for PTX3, CXCR2, and RUNDC3A

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Variables of interest

- **Genes:** *PTX3* (from Submission 1), *CXCR2*, and *RUNDC3A*
- **Continuous covariate:** maternal baseline 25-hydroxyvitamin D (ng/mL)
- **Categorical covariate 1:** pregnancy outcome (Normal vs Preeclampsia)
- **Categorical covariate 2 (derived from continuous):** vitamin D status (Deficient < 20, Insufficient 20-29.9, Sufficient 30+ ng/mL)

```
# load needed library
library(tidyverse)
```

1. Load the raw files

```
covar     <- read_csv("data/GSE85307_covar-1.csv")
expr      <- read_csv("data/GSE85307_geneExpression.csv")
geneNames <- read_csv("data/GSE85307_geneNames.csv")
```

2. Clean the covariate data (same steps as Submission 1)

Each covariate value is stored as text such as `"bmi_kg/m2: 37.52"`, so keep only the part after `": "` and convert numbers to numeric.

```
covar_clean <- covar |>
  # Rename first column name by position
  rename(subjectID = 1) |>
  # For every column except the IDs, keep only the text after ": "; consulted
  # the help of Claude for this step to help with str_removal, detecting patterns
  mutate(across(-c(subjectID, geoID),
                ~ str_remove(.x, "^.*: "))) |>
  # Convert the continuous variables from text to numeric
  mutate(across(c(mat_age_yrs,
                  delivery_gestAge_wks,
                  sampleCollection_gestAge_wks,
                  mat_BMI,
                  mat_vitD),
                as.numeric)) |>
  mutate(
    # Pregnancy outcome as a factor, with normal and preeclampsia levels
    preg_preeclampsia = factor(preg_preeclampsia,
                               levels = c("Normal", "Preeclampsia")),
    # Turn asthma into a labeled factor, either 0 or 1
```

```
    mat_asthma = factor(mat_asthma,
                        levels = c("0", "1"),
                        labels = c("Healthy", "Asthma")),
    # convert to factor variable
    batch_Microarray = factor(batch_Microarray),
    # turn continuous mat_vitD to categorical vitD_status variable
    # chose the intervals and cutoffs based off of existing clinical studies
    # and standard medical practices
    vitD_status = cut(mat_vitD,
                      breaks = c(-Inf, 20, 30, Inf),
                      right  = FALSE,
                      labels = c("Deficient (<20)",
                                 "Insufficient (20-29.9)",
                                 "Sufficient (30+)"))
  )

summary(covar_clean)
count(covar_clean, preg_preeclampsia, vitD_status)
```

3. Build a long data frame with all of the genes

Building a data frame that holds every gene, because the function needs a data frame, so it can pull out whichever genes it is given.

```
expr_long <- expr |>
  # Probe ID column has no header; name it so it can be joined
  rename(PROBEID = 1) |>
  # Attach gene names from the linker file
  left_join(geneNames, by = "PROBEID") |>
  # convert to long format, so one row per gene per sample
  pivot_longer(cols = starts_with("GSM"),
               names_to  = "geoID",
               values_to = "expression")

# double check that every probe has a gene name, and no gene appears twice
sum(is.na(expr_long$GENE))
expr_long |>
  distinct(PROBEID, GENE) |>
  count(GENE) |>
  filter(n > 1)  |>
  nrow()
```

4. Merge with covariates and verify the match

```
# check if set of sample IDs in both files?
setequal(unique(expr_long$geoID), covar_clean$geoID)

analysis_long <- covar_clean |>
  inner_join(expr_long, by = "geoID")

# Expect 157 samples x 500 genes = 78,500 rows, with 157 unique samples
# passed the check
nrow(analysis_long)
```

```r
n_distinct(analysis_long$geoID)

# Check if merged PTX3 value for one sample equals the raw file value
analysis_long |>
  filter(geoID == "GSM2264346", GENE == "PTX3")  |>
  pull(expression)

expr |>
  rename(PROBEID = 1)  |>
  filter(PROBEID == 8083594)  |>
  pull(GSM2264346)
```

5. Shared plot theme and colors (same from Submission 1)

```r
# One theme for consistency across all plots; publication quality
theme_pub <- theme_classic(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5),
        axis.title = element_text(face = "bold"),
        legend.position = "top",
        legend.title = element_text(face = "bold"))

# shared colors
outcome_colors <- c("Normal" = "#4C72B0", "Preeclampsia" = "#C44E52")
```

6. Plotting function

plot_genes() function takes the following inputs: a data frame and a vector of one or more gene names. For each gene it loops through, it makes the three Submission 1 plots (histogram, scatterplot vs vitamin D, boxplot by vitamin D status and outcome) and prints them, optionally saves them, and returns them in a named list.

```r
# takes the input of a data frame, a vector of genes, into the figures directory
plot_genes <- function(df, genes, save = FALSE, out_dir = "figures") {

  # First initialize an empty list
  all_plots <- list()

  # Loop through each gene provided and produce the plots (huge for loop!)
  for (gene in genes) {

    # Keep only this gene's rows (one row per sample)
    # rename input data frame as gene_df
    gene_df <- df |>
      filter(GENE == gene)
    # Axis label reused by all three plots
    y_lab <- paste(gene, "expression (log2 intensity)")

    # Plot 1: histogram showing distribution of expression; essentially same
    # code as submission 1
    p_hist <- ggplot(gene_df, aes(x = expression)) +
      # 0.05-wide bins suit the narrow range of log2 expression values
      geom_histogram(binwidth = 0.05, fill = "#3A7D7C", color = "white",
                     linewidth = 0.3) +
```

```
  # Dashed line at the median as a reference point
  geom_vline(aes(xintercept = median(expression)),
             linetype = "dashed", color = "#1F2A37", linewidth = 0.6) +
  # Bars sit directly on the x-axis, with some space on top
  scale_y_continuous(expand = expansion(mult = c(0, 0.05))) +
  labs(title = paste("Distribution of", gene, "Expression"),
       x = y_lab,
       y = "Number of samples",
       caption = paste0("Dashed line = median; n = ", nrow(gene_df))) +
  theme_pub

# Once again same thing but for plot 2: scatterplot of vitamin D
# vs expression. same code as submission 1, just changing the inputs.
p_scatter <- ggplot(gene_df, aes(x = mat_vitD, y = expression)) +
  # One point per woman, colored by outcome
  geom_point(aes(color = preg_preeclampsia), alpha = 0.7, size = 2.5) +
  # Overall linear trend with 95% confidence band
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  scale_color_manual(values = outcome_colors) +
  labs(title = paste("Maternal Vitamin D vs", gene, "Expression"),
       x = "Maternal 25(OH)D at baseline (ng/mL)",
       y = y_lab,
       color = "Pregnancy outcome") +
  theme_pub

# Same thing again, but for plot 3: boxplot by both categorical variables
p_box <- ggplot(gene_df, aes(x = vitD_status, y = expression,
                             fill = preg_preeclampsia)) +
  # Boxes side by side for each outcome; hide outliers because
  # all points are drawn below
  geom_boxplot(outlier.shape = NA, alpha = 0.8,
               position = position_dodge(width = 0.8)) +
  # Individual women, jittered sideways and aligned with their box
  geom_point(position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width  = 0.8),
             size = 1.2, alpha = 0.6) +
  scale_fill_manual(values = outcome_colors) +
  # edit in that particular looped gene
  labs(title = paste(gene, "Expression by Vitamin D Status and Outcome"),
       x = "Maternal vitamin D status (ng/mL)",
       y = y_lab,
       fill = "Pregnancy outcome") +
  theme_pub

# Print all three plots so they appear in the knitted document
print(p_hist)
print(p_scatter)
print(p_box)

# Save the plots named by gene if save is set to TRUE, the default input
# is that save is set as FALSE initially; this is optional, that way doesn't
# continuously rewrite plots each time functino is run
if (save) {
```

```
      ggsave(file.path(out_dir, paste0("histogram_", gene, ".png")),
             p_hist, width = 7, height = 5, dpi = 300)
      ggsave(file.path(out_dir, paste0("scatter_vitD_", gene, ".png")),
             p_scatter, width = 7, height = 5, dpi = 300)
      ggsave(file.path(out_dir, paste0("boxplot_vitD_outcome_", gene, ".png")),
             p_box, width = 8, height = 5, dpi = 300)
    }

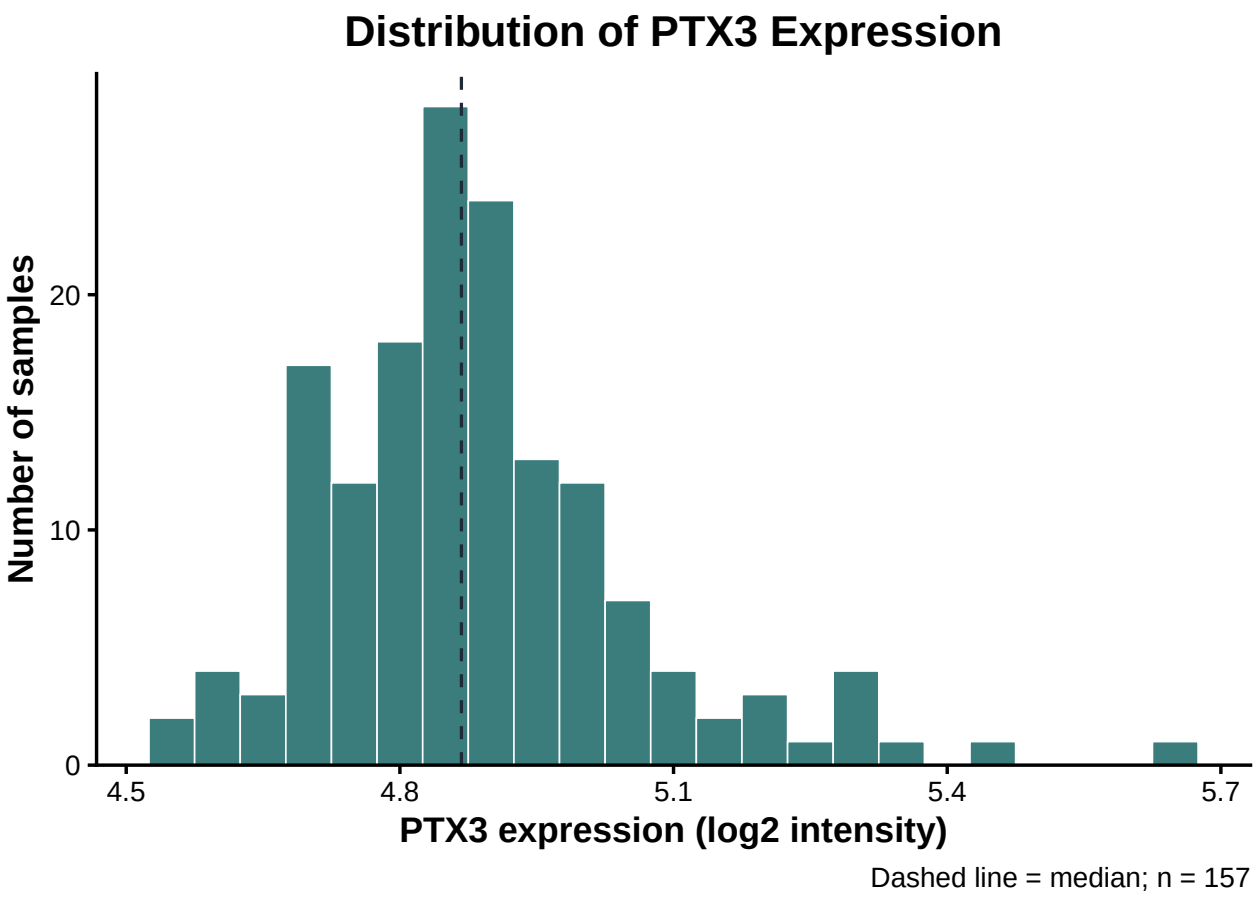
    # Store this gene's plots in the list under the gene's name
    # This way when the loop finishes you can pull out any single plot later
    # without rerunning the function
    all_plots[[gene]] <- list(histogram = p_hist,
                              scatter   = p_scatter,
                              boxplot   = p_box)
  }
}
```

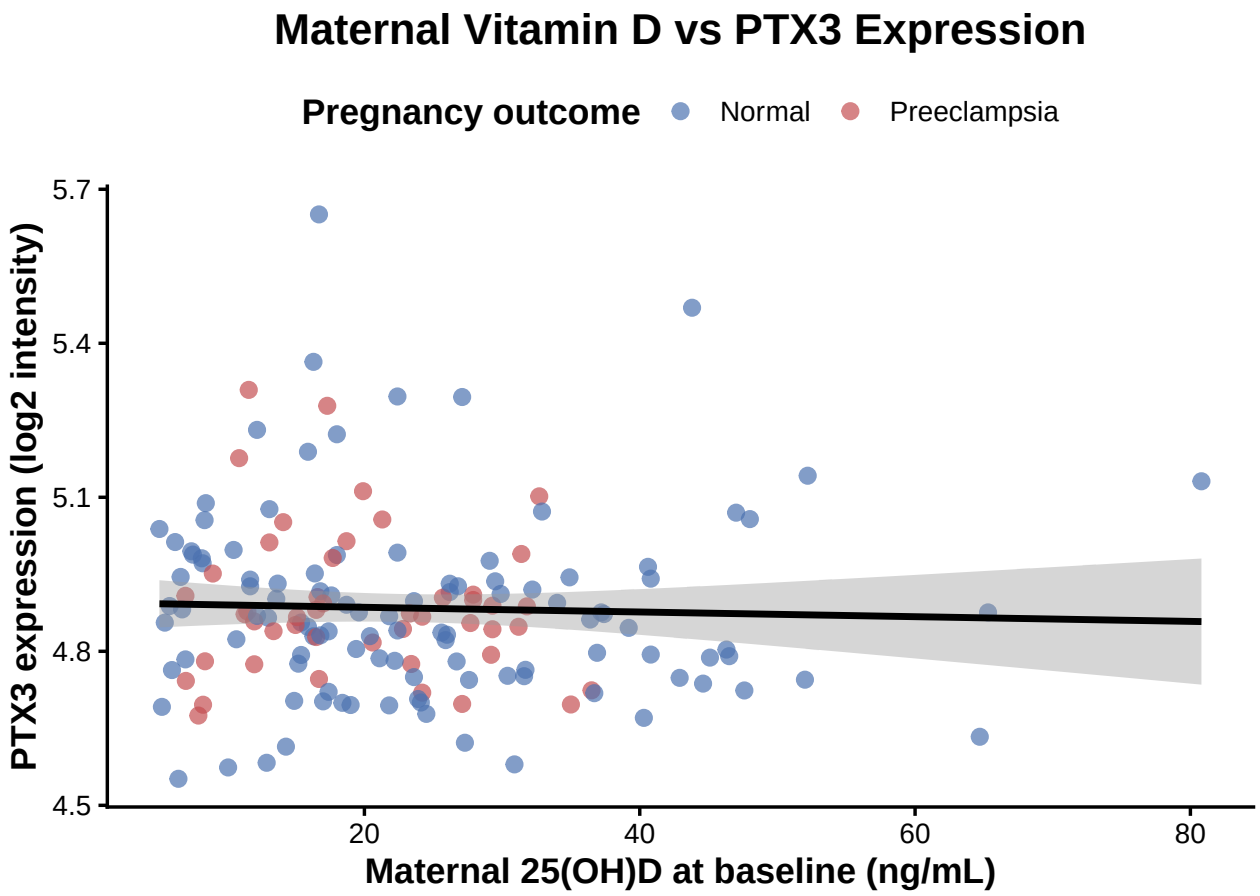
7. Run the function on three genes

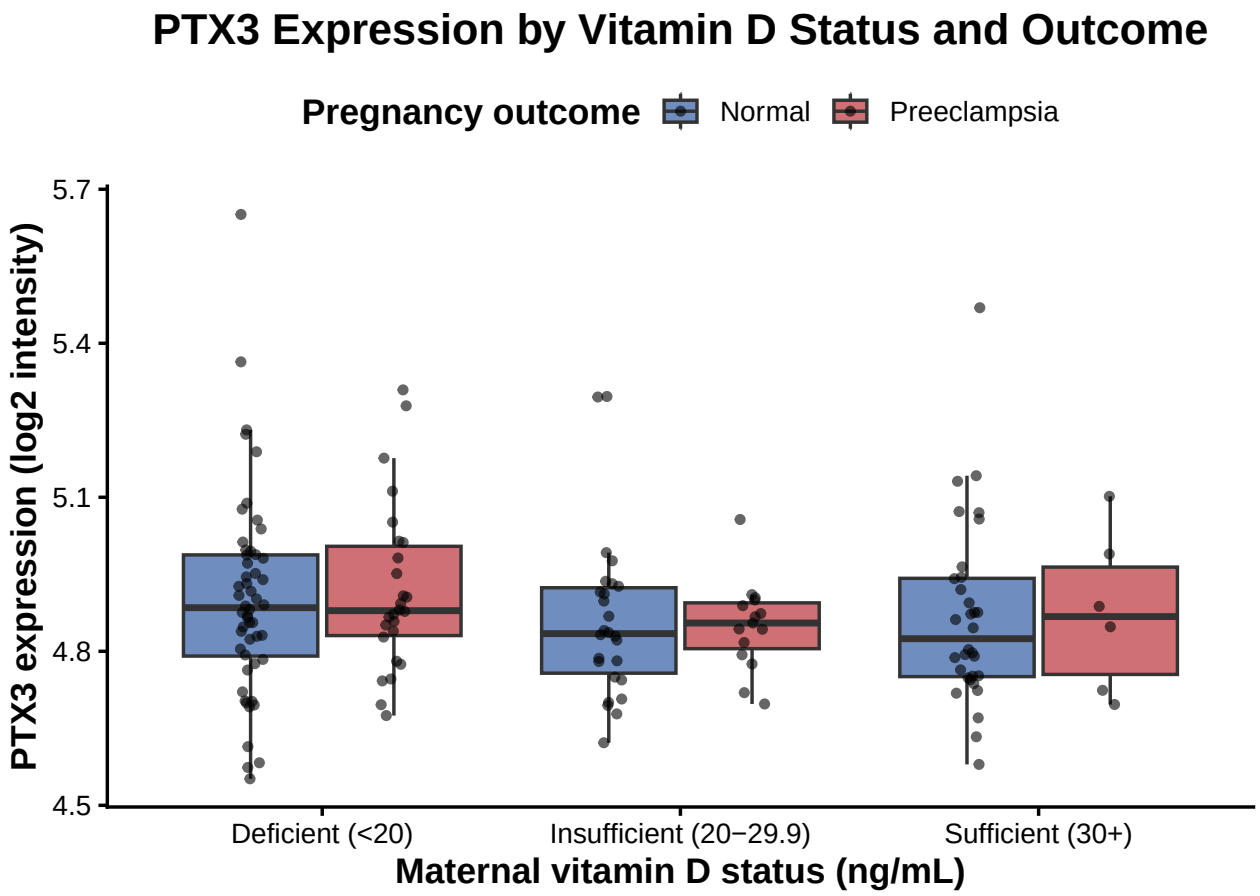
```
# interested genes in the list
genes_of_interest <- c("PTX3", "CXCR2", "RUNDC3A")

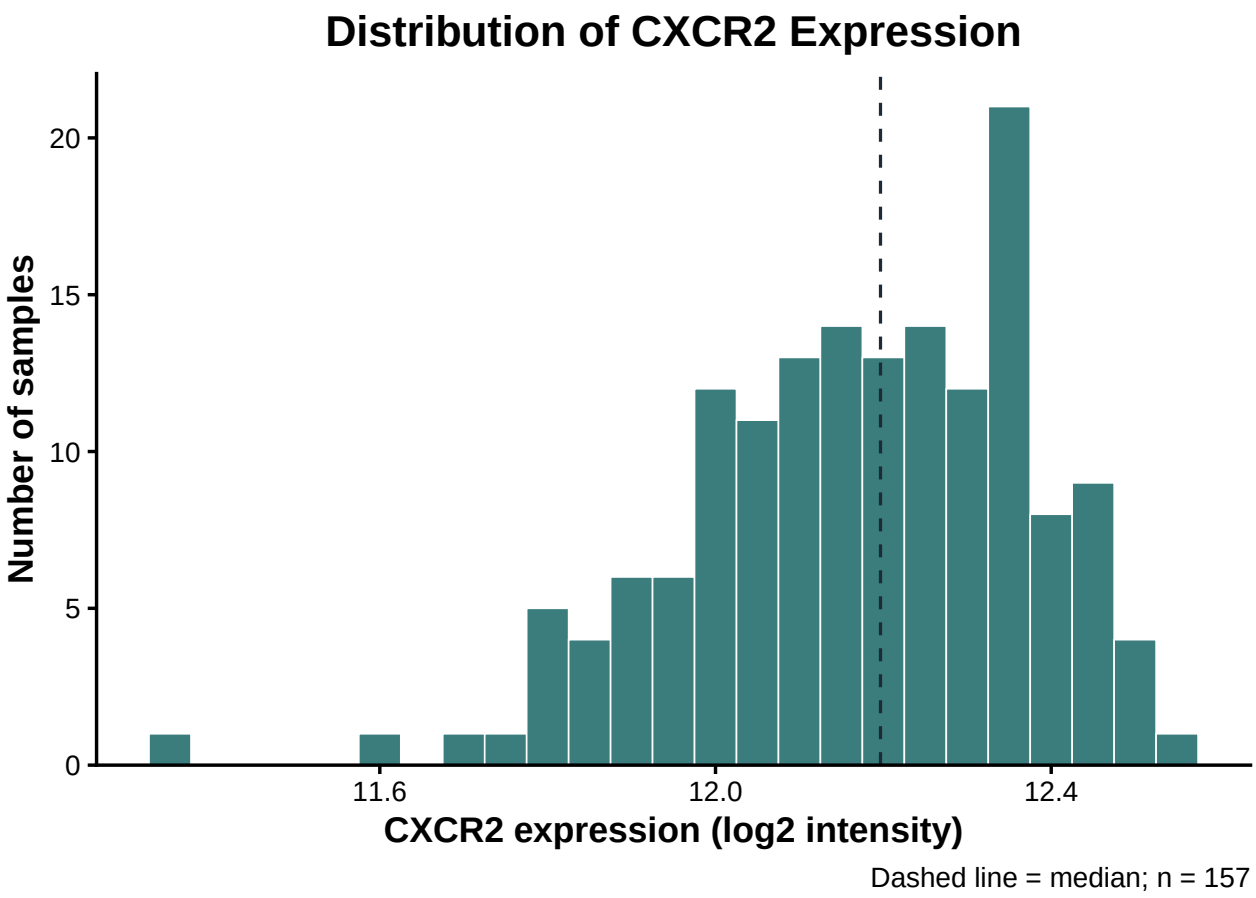
# Set save = TRUE to save the 9 figures in figures folder
# Set save = FALSE in final pdf/markdown because don't want to keep running
# this over and over

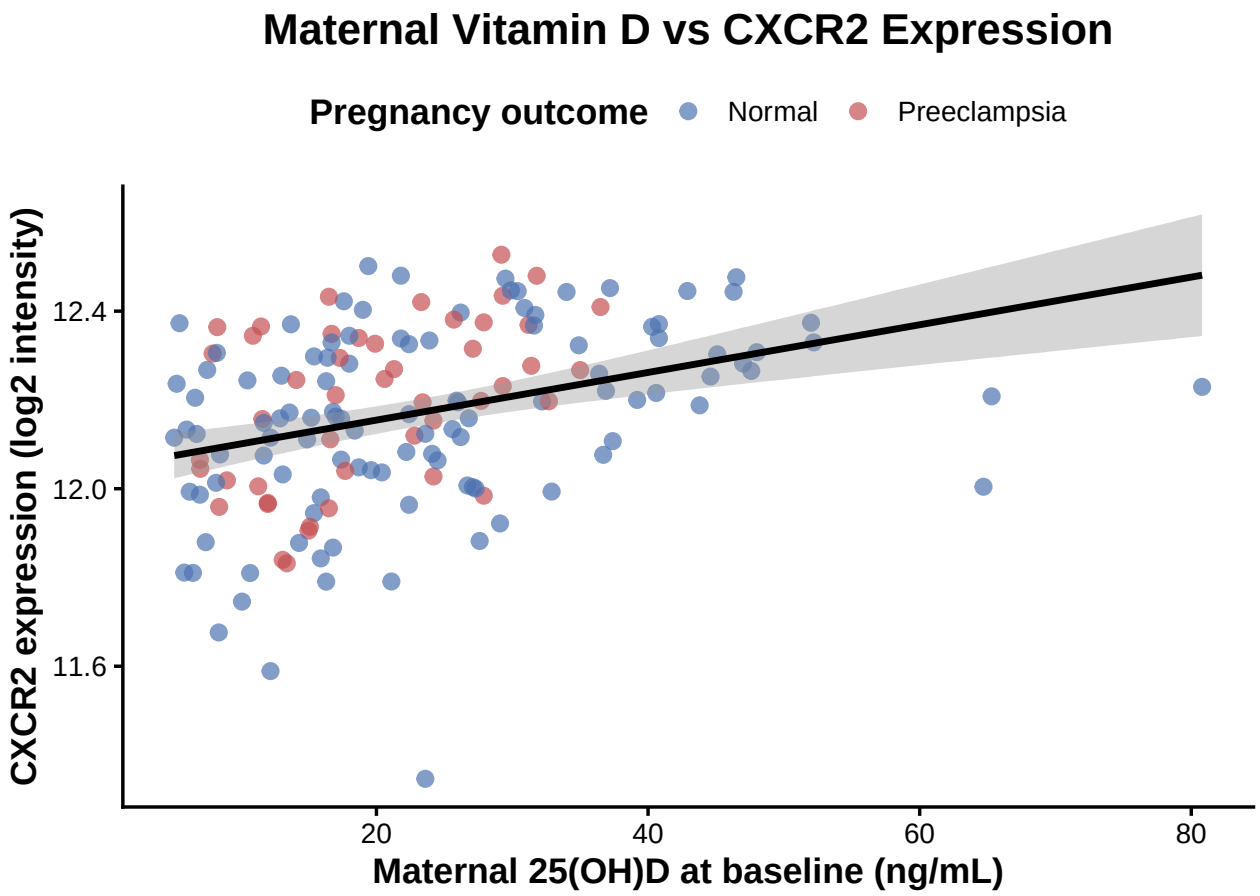
# takes the analysis_long data frame input created from step 3 and 4 as input
# also takes list genes in previous line as input
gene_plots <- plot_genes(analysis_long, genes_of_interest, save = FALSE)
```

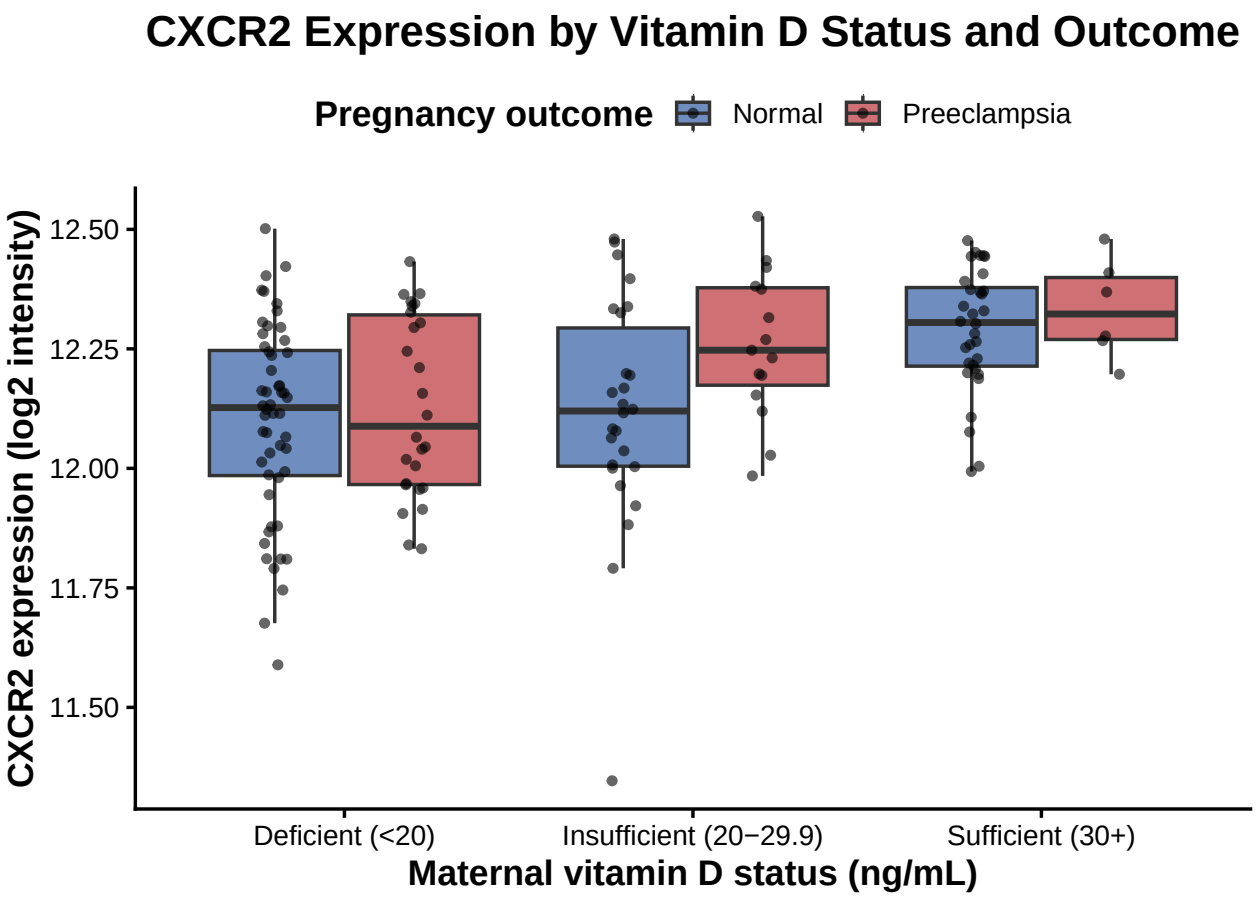

Distribution of PTX3 Expression
Number of samples
0
10
20
4.5
4.8
5.1
5.4
5.7
PTX3 expression (log2 intensity)
Dashed line = median; n = 157

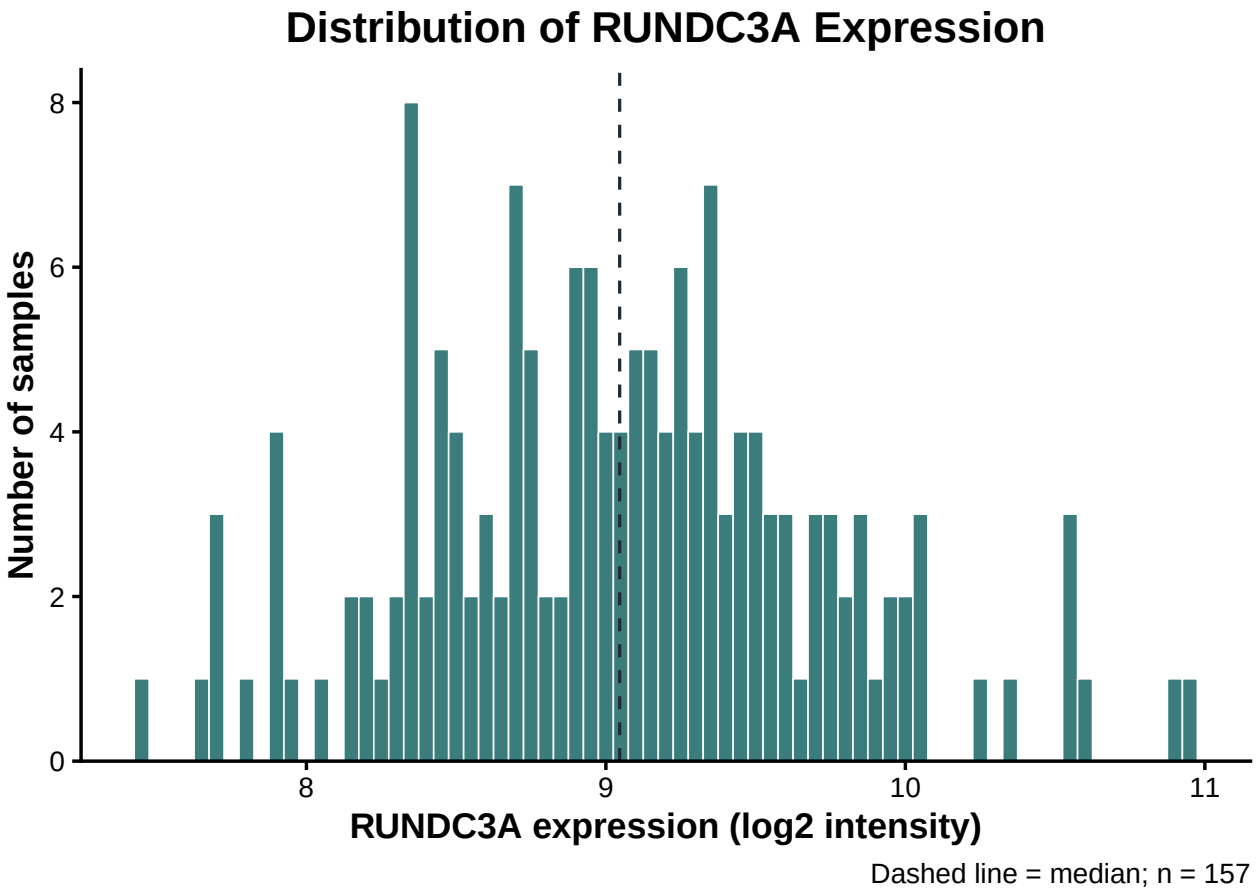

Maternal Vitamin D vs PTX3 Expression
Pregnancy outcome
Normal
Preeclampsia
PTX3 expression (log2 intensity)
5.7
5.4
5.1
4.8
4.5
20
40
60
80
Maternal 25(OH)D at baseline (ng/mL)

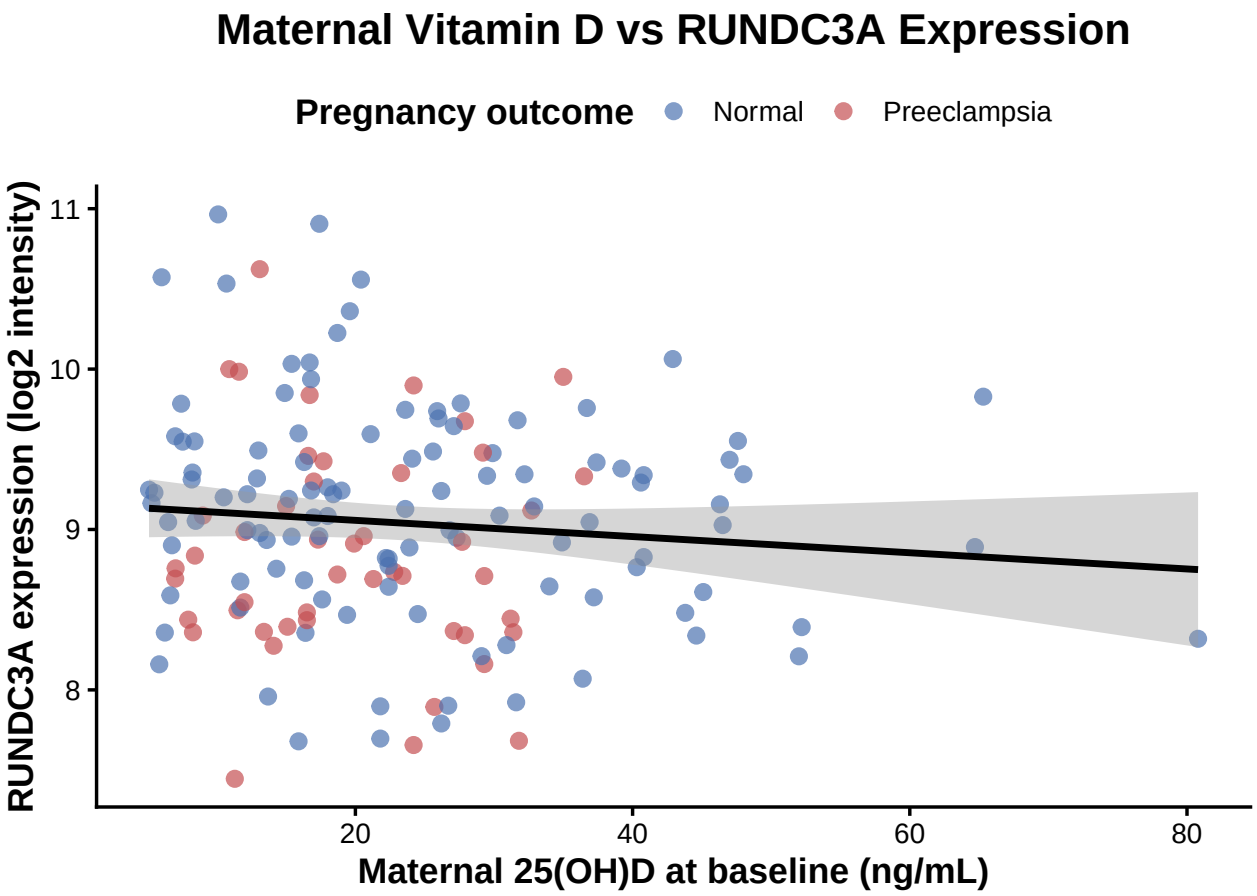

PTX3 Expression by Vitamin D Status and Outcome
Pregnancy outcome
Normal
Preeclampsia
PTX3 expression (log2 intensity)
5.7
5.4
5.1
4.8
4.5
Deficient (<20)
Insufficient (20−29.9)
Sufficient (30+)
Maternal vitamin D status (ng/mL)

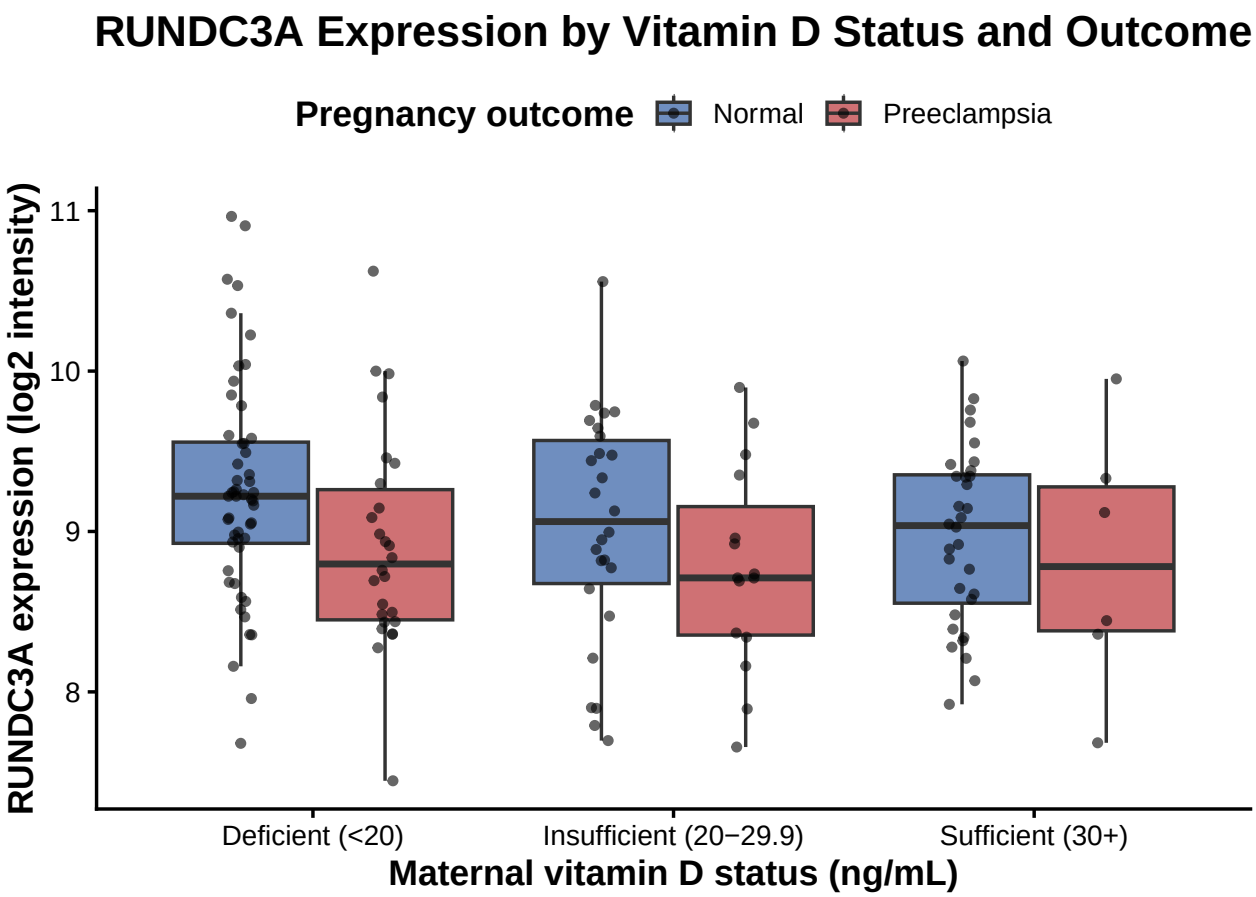

Distribution of CXCR2 Expression
Number of samples
0
5
10
15
20
11.6
12.0
12.4
CXCR2 expression (log2 intensity)
Dashed line = median; n = 157


Maternal Vitamin D vs CXCR2 Expression
Pregnancy outcome
Normal
Preeclampsia
CXCR2 expression (log2 intensity)
12.4
12.0
11.6
20
40
60
80
Maternal 25(OH)D at baseline (ng/mL)


CXCR2 Expression by Vitamin D Status and Outcome
Pregnancy outcome
Normal
Preeclampsia
CXCR2 expression (log2 intensity)
12.50
12.25
12.00
11.75
11.50
Deficient (<20)
Insufficient (20−29.9)
Sufficient (30+)
Maternal vitamin D status (ng/mL)


Distribution of RUNDC3A Expression
Number of samples
0
2
4
6
8
8
9
10
11
RUNDC3A expression (log2 intensity)
Dashed line = median; n = 157


Maternal Vitamin D vs RUNDC3A Expression
Pregnancy outcome
Normal
Preeclampsia
RUNDC3A expression (log2 intensity)
11
10
9
8
20
40
60
80
Maternal 25(OH)D at baseline (ng/mL)



AI use statement: Claude (Anthropic) was used to help with str_removal for initial data cleaning and some GitHub debugging issues for function creation. Also consulted Claude for help specifically with including the ggsave portion in the function writing, and consulted google as well for ggsave arguments within the function. Most of the code remained the same as submission 1.